



Abertay University

School of Design and Informatics

HCS509: Human Factors

Unit 1 Portfolio

Learning Outcomes Assessed:

- LO1.** Critically appraise Human Factors and User experience in relation to computer systems
- LO2.** Apply state of the art techniques in assessing and creating Human Factor concepts in relation to interfaces within the context of computer systems.
- LO3.** Design a Human Factors experiment to investigate human computer interaction in relation to a computer system and utilise processes for assessing usability and accessibility.

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Human Factors Portfolio

This assessment brief outlines everything that is required to create your assessed portfolio for HSC509: Human Factors. The creation of your portfolio will require you to demonstrate an understanding of each of the concepts covered throughout this module.

Your objective is to conceive of your own idea for a new website, mobile app, or other type of technological tool and take this idea through the earliest stages of development. For the sake of simplicity, the idea that you have chosen to develop will be referred to as your product.

The early development of your product will involve:

- Researching the market to understand the potential user base and creating a persona
- Analysing the strengths and weaknesses of similar existing products
- Designing a Low-Fidelity prototype of your product based on your research
- Creating a research plan for how you might gain feedback on your product
- Writing a report describing your journey through this process

To keep the development of your portfolio on track, this assessment has been broken down into two parts with distinct stages. Each stage has an objective section describing the focus, a mission section detailing what you should accomplish, and an artifacts section stating what should be submitted to demonstrate your accomplishments.

It is advised that you begin creating your portfolio during week two of this module and progress through each of the stages below as the topics are covered in the course material.

Portfolio Assessment - Part 1

Part 1 should contain all artefacts for stage 1 and should be submitted by **12PM on Monday of week 4**.

Stage 1: Defining Your Product and Persona Development

Whether your product is a website, a mobile app or something else, you will be expected to follow a user-centered design process in these early stages of development.

Objective

Once you have a good idea of the type of product you wish to develop, you should first seek to gain an understanding of its potential user base.

Mission

Consider the high-level objectives you want the product to achieve. Does your product convey information? Does it entertain? Does it provide a service? Once you answered these questions you will have a good idea of who your products potential users are. You should then conduct research on those users to create a user persona. You may conduct your research in any manner available to you using online forums, statistics from journal or credible news articles, or even through the conduction of interviews. However, all aspects of your persona must be fully justified.

Artifacts

1: A 500-word (+-10%) report providing a brief explanation of your product and outlining your research that justifies the features of your persona. It is vital that you provide at least one reference which justifies each feature other than the persona name.

2: A persona with the following features:

- Persona Name - Give your persona a name for easier reference.
- Persona Image - Using a copyright free stock image to indicate what the average user might look like.
- Demographics - Include age, location, occupation. etc.
- Background - Describe their experience using similar products, whether they are experts at using products like yours or novices.
- Goals - List their primary objectives when using a product like yours, what do they want to get out of it?
- Challenges - Identify any obstacles the typical user might have when using a product like yours.
- Motivations - Outline what compels them to user a product like yours and what brings them back to it.
- Tech Savviness - Describe their comfort with technology, would they instinctively know how to use a product like yours?
- Safety Concerns - Highlight any specific safety concerns related to products like yours. If it's a website or mobile app this might be viruses.
- Product Expectations - List the features or information they expect from a product like yours.

Portfolio Assessment - Part 2

Part 2 should contain all artefacts for stages 2, 3, and 4 and should be submitted by **12PM Monday of week 8**.

Final Report

While completing the next stages within this part you should begin a final report that introduces your product and contains a final reflection on what you have accomplished.

Stage 2: Analysis of Existing Products and User Story Creation

Now that you have created your persona you are ready to identify how potential users interact with existing products like yours and possibly gain insights that might help improve your product.

Objective

There are likely to be products that already exist out there which are similar to yours. By conducting an analysis of these products, you will be able to understand what makes them popular and even spot potential flaws.

Mission

Select an existing product that is similar to your own and conduct a usability analysis. Your objective is to uncover both the strengths and weaknesses of the product. You should keep in mind how your user persona might interact with the product.

Artifacts

1: Add a section to your final report titled “Stage 2” which is **700** -words (+-10%) and contains the following:

- **The product** - Describe the product you have chosen to analyse. Try to identify what the developers of the chosen product were thinking during its creation, what purpose does it have, and why is it similar to your product.
- **Identifying Strengths and Weaknesses** - This section should list of the chosen products strengths and weaknesses. Focus on aspects like usability, navigation, information layout, interactivity, and overall user experience. Describe the marketing strategy of the product, is it successful in engaging with the users?
- **Design Elements** - This section should note the products design elements such as colour schemes, typography, button placement, and overall aesthetics. Describe how these elements contribute to or hinder usability and overall user experience.
- **Areas for Optimization** - In this section you should identify at least three areas where the product you are examining could be improved.

Stage 3: Low-Fidelity Prototyping

By this point you will have an idea of your target user, the similar existing products, and how your target user group might interact with them. It is now time to put all of the information you have gathered into further developing your own idea.

Objective

This stage will turn your idea into something physical.

Mission

Up until now your idea will have existed within your own mind. In this stage you will create several artifacts that will allow you to better communicate what your product is. This is vital, especially when working with a team who will develop your product or attempting to convince investors to back the creation of your product.

Use pencil and paper or your choice of digital drawing tools (Microsoft Paint, Adobe XD Free Version, or any online sketching tools) to create the artefacts for this stage.

Artifacts

- 1:** Add a section to your final report titled “Stage 3” which is **500** -words (+-10%) and contains the following: What you learned from this prototyping stage. What worked well? What would you change in your next iteration? Document any usability concerns you might have discovered at this stage, consider navigation ease, content clarity, and aesthetic appeal.
- 2:** Design Sketches - Create two different design ideas for your product. Focus on main features, layout and colour scheme.
- 3:** A Low-Fidelity Prototype - Chose one of your design and create a low-fidelity prototype of your product based on your sketches.
- 4:** A User Storyboard - Outline how the user persona you designed might use your product.
- 5:** A User Flow Diagram - This diagram should show each how the average user will navigate your product.

Stage 4: Research methodology proposal

In this final stage you will create a plan to expose a hypothetical prototype of your product to potential users and gather data on their reaction.

Objective

Identifying areas for improvement or potential user concerns at the prototyping stage can save a vast amount of time and funds. This stage will focus on creating a plan for gathering feedback.

Mission

By now you will have learned about the differences between qualitative and quantitative research. With these in mind you should develop a research plan which will capture useful data regarding how real-world users might interact with your product.

Artifacts

1: Add a section to your final report titled “Stage 4” which is **800-words** (+-10%) and contains the following:

- Objective - Clearly state the aim or goal of the usability test.
- Data Collection Strategy - Outline how you would gather data, which could include user interviews, think-aloud sessions, or pre-and-post interaction surveys. State which research method and what type of data you are collecting, quantitative or qualitative or mixed?
- Participant Strategy: Formulate how you would select and reach out to test participants. Draft a test script outlining the procedure, questions, and user interactions you would observe. What types of questions do you want the user to answer. Come up with 5-6 questions you think would be important to learn to help you in redesigning the site.
- Data Analysis - Outline how you would analyse the data you collected, and how you would use this to inform and update your products design.

Submission Instructions

All reports should be submitted in Word or PDF format and named (“Stage number” Report “your student number”).

All other artifacts should be submitted as files alongside reports. **Do not use .zip files.**

NOTE: that if you have taken photos or provided sketches as artifacts these should all be high-resolution, clearly visible, and readable.

Grades

Please consult the rubric for this practical assessment on canvas, a link to the rubric will appear alongside the download option for this brief. Your portfolio contributes 100% of the final grade for this module. Grades and feedback will be provided 15 working days after the submission deadline.

Mitigating Circumstances

If you have Mitigating Circumstances (see mitigating circumstances policy) that prevent you from completing or submitting your work on or before the deadline, you should contact the Module Leader before the submission date. Your Module Leader will advise if extra time is possible or direct you to the self-certification procedure if applicable.

Late Submission

Submissions made after their deadlines (excluding those with mitigating circumstances) will be deemed as a late submission and will incur a penalty, including the possibility of the work being awarded a non-submission (NS) grade.

Late coursework will be adjusted on a sliding scale with one grade deducted (e.g. from B to C+) for each day late for the first two days, after which the grade awarded will be NS.

Plagiarism

Please see the university regulations on plagiarism. Students are encouraged to have discussions; however, only original work should be submitted. Plagiarism may result in a zero mark being recorded for the coursework and may result in further action being taken. Please see the University regulations, codes & policies:

<https://intranet.abertay.ac.uk/library/referencing/avoidingplagiarism/>

Use of AI in this Module

Please Note: The stipulations below apply only to this module, you will normally be given specific guidance regarding Generative AI for each module that you study and should not assume that the use of AI is allowed if it is not specified.

You may use Generative AI to:

- a. understand concepts or questions related to lectures, tutorials, practicals, or the assessments for this module.
- b. get suggestions for refining ideas, report structure, proofreading, aiding grammar and spelling.
- d. generate images for use in assessments

You may NOT use Generative AI to:

- a) generate content for reports.
- b) generate content for the discussion board.
- c) generate answers to questions posed in briefs or other text for your reports.

Any evidence of AI use outside or contrary to the guidelines above will be referred to the Student Disciplinary Officer within the School and may result in Academic Misconduct charge.

You may use specific tools, such as Grammarly for aiding Grammar and Structure.

- a) It is recommended that you keep a record of your interaction with these tools.
- b) It is recommended that you have evidence of drafts of your work.

You should adhere to the following:

1. Always reference AI use: Consider AI like any other source. If AI helps draft or edit a report or other work, reference it appropriately.
2. Record Interaction: When using ChatGPT or a similar tool, include a transcript or screenshots of your interaction in an appendix (for reports).
3. Summary: Include a summary in an appendix stating how you used the AI.
4. In-text Citation for AI: For each paragraph aided by AI, include an in-text citation.
5. Own thoughts: The majority of words, insights and arguments in your work should be your own. Using AI for proofreading or structure is acceptable, but the core ideas must be original.
6. Be Responsible: Use AI responsibly to enhance your work. Critical thinking and creativity should come from you. Speak with lecturer if you are unsure.

Additional Points to Consider:

1. Ethical AI Usage: When using AI in subject ensure the data and methods you employ adhere to privacy laws and ethical considerations.
2. Validity of Data: When using AI for data generation or simulation, clearly state the limitations and potential biases of the model in your report.
3. Transparency in Team Projects: In the case of team projects, disclose the extent of AI use to your team members.
4. Factual Accuracy: Consider the information generated by AI is not factually accurate and or scientifically valid. The information generated should be checked against peer-reviewed sources.